

# JEE MAIN REVISION MASTER: SEMICONDUCTOR DEVICES AND LOGIC GATES 2026 3 QUESTIONS

## 5% WEIGHTAGE $\uparrow$ 50.15%

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 32 Analyzed PYQs

### 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                                | Frequency | Category             |
|---------------|---------------------------------------------|-----------|----------------------|
| Very High     | p-n Junction Diode & Zener Diode Regulation | 19 PYQs   | Semiconductor Diodes |
| High          | Logic Gates & Truth Tables                  | 13 PYQs   | Digital Electronics  |

### 2. Master Formula Reference Card

| Concept / Topic                             | Mathematical Formulation & Governing Equations                                                                                                                                               |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| p-n Junction Diode & Zener Diode Regulation | $\eta_{\text{half-wave}} = 40.6\%$ , $\eta_{\text{full-wave}} = 81.2\%$ , $V_L = V_Z = \text{const}$ , $I_S = I_Z + I_L$                                                                     |
| Logic Gates & Truth Tables                  | $\text{NAND: } Y = \overline{A \cdot B}$ , $\text{NOR: } Y = \overline{A + B}$ , $\overline{A \cdot B} = \overline{A} + \overline{B}$ , $\overline{A + B} = \overline{A} \cdot \overline{B}$ |

### 3. Deep-Dive Concepts & Linked PYQ Evidence

#### 3.1 p-n Junction Diode & Zener Diode Regulation (Very High Yield — 19 PYQs)

**Core Principle:** Forward and reverse characteristics of p-n junctions, rectifiers, ripple frequency, and Zener diode shunt voltage stabilization.

■ **Caution / JEE Pitfall:** In full-wave rectifiers, output ripple frequency is TWICE the AC input frequency ( $2f_{in}$ ), whereas for half-wave it is  $f_{in}$ .

■ **Speed Trick:** In reverse breakdown, Zener diode maintains constant voltage  $V_Z$  across its terminals as long as  $I_Z \geq I_{Z,min}$ .

**Representative Source PYQs:**

- [2026 Shift] Q#1 (MCQ): "A wafer of pure germanium crystal has two parts X and Y. The end X is obtained by doping with arsenic and Y with indium. It is connected to ..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "In which of the following figures, diode is reverse biased?..." [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "The circuit shown in figure contains two ideal diodes  $D_1$  and  $D_2$ . If a cell of emf 3V and negligible internal resistance is conn..." [Ans: N/A]

## 3.2 Logic Gates & Truth Tables (High Yield — 13 PYQs)

**Core Principle:** Universal logic gates (NAND and NOR), De Morgan's theorems, and Boolean circuit reduction.

■ **Caution / JEE Pitfall:** NAND and NOR gates are UNIVERSAL gates: any Boolean function can be constructed using only NAND or only NOR gates.

■ **Speed Trick:** Connect inputs of a NAND or NOR gate together to obtain a NOT gate:  $\overline{A \cdot A} = \overline{A}$ .

**Representative Source PYQs:**

- [2026 Shift] Q#3 (MCQ): "An n-type and p-type semiconductor can be obtained by respectively doping pure silicon with..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "The upper level of valence band and lower level of conduction band overlap in the case of..." [Ans: N/A]
- [2023 Shift] Q#4 (MCQ): "A truth table for the given circuit is..." [Ans: C]

## 4. Benchmark Illustrative PYQs with Master Solutions

### Benchmark Problem 1: KCET 2026 (Q#1)

A wafer of pure germanium crystal has two parts X and Y. The end X is obtained by doping with arsenic and Y with indium. It is connected to a battery as shown in the figure. Which of the following statements is correct?

- (A) X is p-type, Y is n-type and the junction is forward biased
- (B) X is n-type, Y is p-type and the junction is forward biased
- (C) X is p-type, Y is n-type and the junction is reverse biased
- (D) X is n-type, Y is p-type and the junction is reverse biased

**Correct Answer: None**

**Solution Analysis:** Part X is doped with arsenic. Arsenic is a Group 15 element, so it has 5 valence electrons. When it is added to germanium, it gives extra free electrons. So, part X becomes an n-type semiconductor. Part Y is doped with indium. Indium is a Group 13 element, so it has 3 valence electrons. When it is added to germanium, it creates holes. So, part Y becomes a p-type semiconductor. In the battery symbol, the longer line represents the positive terminal, and the shorter line represents the negative terminal. From the figure, part X is connected to the positive terminal and part Y is connected to the

### Benchmark Problem 2: KCET 2026 (Q#3)

An n-type and p-type semiconductor can be obtained by respectively doping pure silicon with

- (A) Arsenic and Phosphorous respectively
- (B) Indium and Aluminium respectively
- (C) Phosphorous and Indium respectively
- (D) Aluminium and Boron respectively

**Correct Answer: None**

**Solution Analysis:** An n -type semiconductor is obtained by doping pure silicon with a pentavalent impurity, such as Phosphorus or Arsenic. A p -type semiconductor is obtained by doping pure silicon with a trivalent impurity, such as Boron, Aluminium, or Indium. So, for silicon: n-type → Phosphorus p-type → Indium Hence, the correct option is: Option C: Phosphorous and Indium respectively